

Project Overview The primary focus of this repository is exploring core data science and statistical concepts natively in R.

Featured Implementation: Gaussian Mixture Model (GMM) This repository includes a native R implementation of a Gaussian Mixture Model (GMM) optimized using the Expectation-Maximization (EM) algorithm. The script generates a synthetic 2D dataset with two distinct clusters, initializes the model parameters, and runs the EM steps iteratively to track convergence.

Synthetic Data Generation: Generates 20,000 data points split evenly across two distinct normal distributions. **Custom Multivariate Gaussian PDF:** Computes probability densities natively without external dependencies.

Expectation-Maximization Loop: Manually implements both the E-step (responsibility calculation) and M-step (parameter updates). **Convergence Tracking:** Calculates and records log-likelihood across iterations to track EM behavior. **Visualization:** Automatically exports plots showing the final clustered data and log-likelihood convergence. data-r

Welcome to the data-r repository! This project contains native R implementations and scripts for data analysis, statistical modeling, and machine learning algorithms.

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Getting Started Prerequisites To run the scripts in this repository, you only need a standard installation of R. No external CRAN packages (like mclust or tidyverse) are required.

Execution Run the main script in your R console or via your terminal:

```
source("gmm.R")
```

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Author

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Course: BCA

Project: Week 1 – Gaussian Mixture Model using EM Algorithm

Language: R (Base R)